

0100EST102052003

Theme of Valuation/Answer Key

Theme of evaluation (marks in brackets) and answers of problems/key)

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
SECOND SEMESTER B.TECH DEGREE EXAMINATION(2019 SCHEME), JUNE 2022

Course Code: EST 102

Course Name: PROGRAMMING IN C (Common to all programs)

Duration: 3 Hours

PART A

Answer all Questions. Each question carries 3 Marks

Marks

1. 5 marks

(3)

1. 5 marks.

Part- 3 marks

(3)

3 marks

(3)

ed and exit controlled-1 mark

3 marks

(3)

structured language. The goto statement is discouraged in C, because it alters the sequential flow of the program. Also it makes difficult to trace program flow control.

o.h>

(3)

g.h>

]; // declaration of char array
]; // declaration of char array

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```

// declaration of integer variable
r the first string : ";
str1);
r the second string : ";
str2);
// both the strings using strcmp() function
p(str1,str2);
)
// strings are same");

```

gs are not same");

am- 3 marks

(3)

 $\langle h.o. \rangle$

```
], size, c, location = 0;
    the number of elements in array\n");
    &size);
    %d integers\n", size);
    < size; c++)
        , &array[c]);
    < size; c++)
        > array[location]
    ;
}
```

num element is present at location %d and its value is %d.\n", location+1, array[location]);	
& example- 1.5 marks	(3)
nce & example – 1.5 marks	
ntages 1.5 marks each	(3)
ity, avoid redundancy, manageability, Improves collaboration	
5 marks	(3)
1.5 marks	
arks	(3)
1.5 marks	
PART B	
Answer any one Question from each module. Each question carries 14 Marks	
with example-4 marks	(14)
arks	
marks	
OR	
ark	(6)
r registers in CPU along with functions (5 marks)	
ithm- 7 marks	(8)
Number	
ze Sum to zero	

Number is not zero Get Remainder by Number Mod 10 Add Remainder to Sum Divide Number by 10 Sum	
Processes with explanation - 4 marks. Memory requirements- 3 marks. Integer-2bytes Whole numbers that can have both zero, positive and negative values but no decimal values. Double are used to hold real numbers. The floating-point data type allows the user to type decimal values, which means you can have more precision than float. This is useful in scientific programs. For example, average marks can be 97.665. If we use int data type, it will strip off the decimal part. For example, 97. To print the exact value, we need 'float' data type. Single character. Char consists of a single byte. Double-8 bytes Float-4 bytes Double-8 bytes Integer-2 bytes Short-2 bytes Long-4 bytes Long long-8 bytes Char-1 byte Float-4 bytes Double-8 bytes Long double-16 bytes Complex-8 bytes Void-1 byte Size of data type can be found using sizeof operator. Compiler/os specific....Eg: int is 4 bytes in gcc linux and mark must be given	
Minimum- 7 marks	(7)
OR	
Functions with example-3.5 marks I/O functions with example-3.5 marks	(7)

am- 7 marks	(7)
n-7 Marks ,h> matrix[10][10], transpose[10][10]; he number of rows and columns of a matrix\n"); ", &m, &n); elements of the matrix\n"); m; c++) < n; d++) , &matrix[c][d]); m; c++) < n; d++) [c] = matrix[c][d]; ose of the matrix:\n"); n; c++) < m; d++) t", transpose[c][d]);	(7)
ing functions along with Syntax -7 marks	(7)

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OR	
am-7 Marks	(7)
<pre><h> g.h> r[100], RevStr[100]; len; \n Please Enter any String : "; r); len - 1; i >= 0; i--) RevStr[j++] = Str[i]; [i] = '\0'; \n String after Reversing = %s", RevStr););</pre>	
am-7 marks	(7)
ation -3Marks	(7)

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ad data- 2 Marks	
display data- 2 Marks.	
definition - 1 mark.	(7)
classes (Auto, extern, static, register)- 1 mark each.	
Mark each	
OR	
marks	(7)
marks	
marks	
mark	(7)
am- 6 marks	
am- 7 marks	(7)
with a sample program- 7 marks	(7)
OR	
adding functions with syntax-7 marks	(7)

m- 7 marks n.h>	(7)
<pre>inter : Print the elements of an array in reverse order :\n"); the number of elements to store in the array (max 15) : "); n);); %d number of elements in the array : \n",n); +) (" element - %d : ",i+1); "%d",pt); - 1]; e elements of array in reverse order are :";); i--) element - %d : %d ", i, *pt);</pre> NOTE: Marks may be awarded for programs conveying correct logic. Sample programs are included in scheme.	